State of art seminar

COMPARATIVE STUDY OF FURNACE AND INFRARED HEATING ON PHASE EVOLUTION, POROSITY, AND MECHANICAL PROPERTIES OF BILAYER CU/AL6061 COLD-SPRAY DEPOSITS

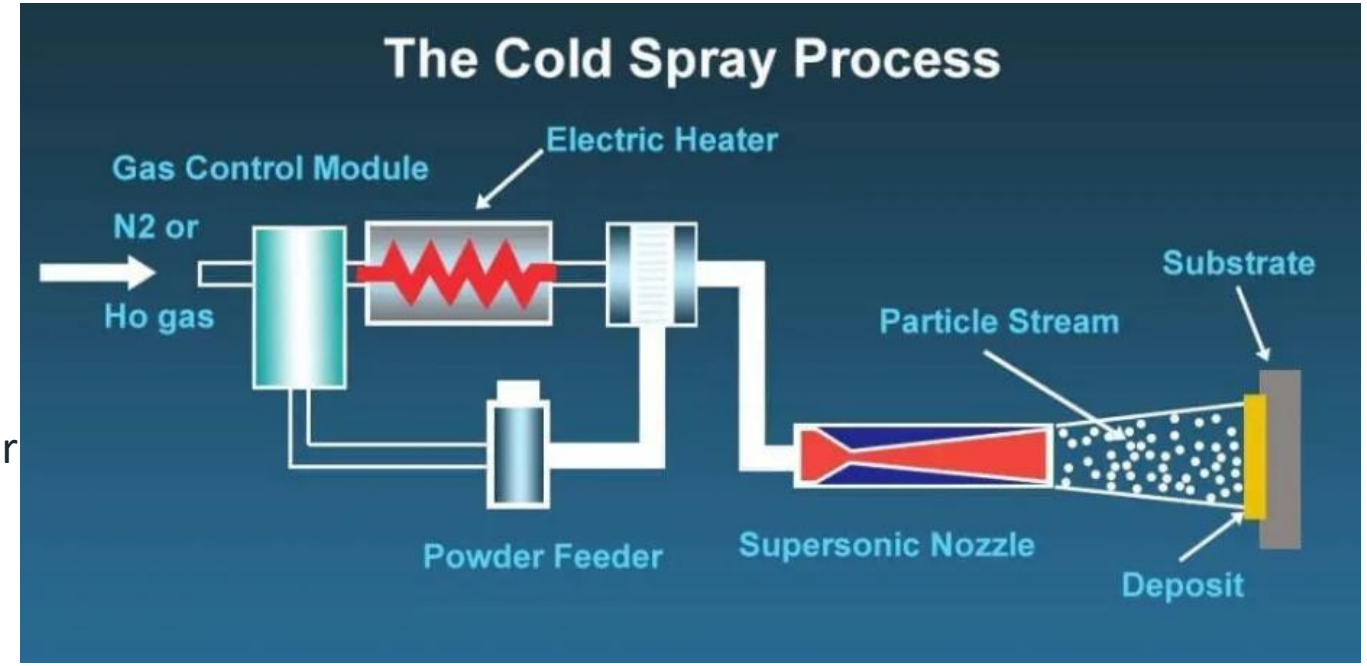
PRESENTED BY:

Dowrla Tanmay Santhosh (2025MEM1023)
Gugulothu Krishna Prasad (2021MEB1284)
Surendra Sahu (2023MEB1388)

Motivation and Problem Definition

What is Cold Spray?

Kinetic energy-driven solid-state deposition particles accelerated to supersonic velocities and plastically deform on impact, building up a dense coating without melting



Why BiLayer Coatings?

- Functional gradient — tailor properties layer by layer
- Improved adhesion between dissimilar materials
- Enhanced wear, corrosion and thermal resistance

Methodology

Step-1: Deposition

Bi-layer coating deposited on Aluminium substrate using Cold Spray.

Step-2: Thermal Treatment

Samples divided into three groups: As-Sprayed, Traditional Heat Treatment and Rapid IR Heating.

Step-3: Phase Identification

XRD performed to identify phase stability and stress relief post-treatment.

Step-4: Mechanical Mapping

Hardness gradient, Porosity analyzed via Vickers, Nano-indentation, Scratch Test across the coating thickness.

Step-5: Microscopy

Cross-sectional analysis using FESEM to evaluate the bi-layer interface and porosity.

Research gap

As-sprayed coatings retain work-hardened, porous microstructures. Post-processing can relax residual stress and densify the coating but the comparative effect of conventional furnace heat treatment (vs) rapid IR heating on bilayer coatings is not well characterized.

Core Objectives

- Evaluate microstructure evolution at the bi-layer interface.
- Compare porosity reduction efficiency: THT vs. IR.
- Characterize phase changes and mechanical gradient.
- Analyze density and hardness recovery postprocessing.

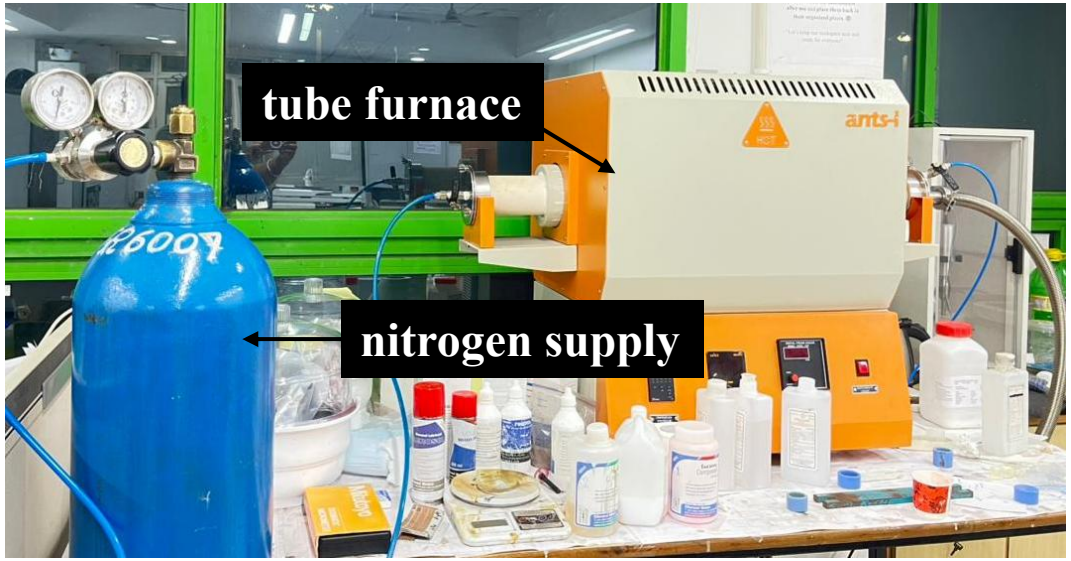
Cold Spray Process Parameters

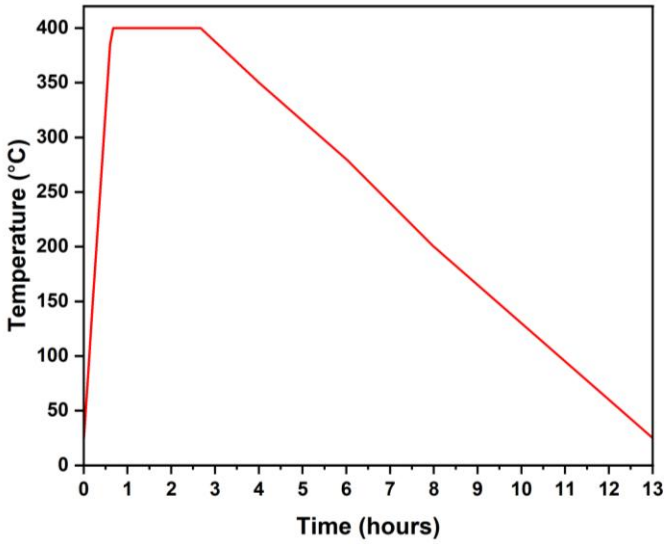
Aluminium Parameters	
Parameter	Value
Process gas	Nitrogen
Pressure	30 bar
Temperature	500 °C
Traverse scanning speed	200 mm/sec
Stand-off distance	30 mm
Number of Passes	11
Step Size	3 mm
Nozzle angle	90°
Feed rate	3RPM

Copper Parameters	
Parameter	Value
Process gas	Nitrogen
Pressure	30 bar
Temperature	700 °C
Traverse scanning speed	50 mm/sec
Stand-off distance	40 mm
Number of Passes	11
Step Size	3 mm
Nozzle angle	90°
Feed rate	3RPM

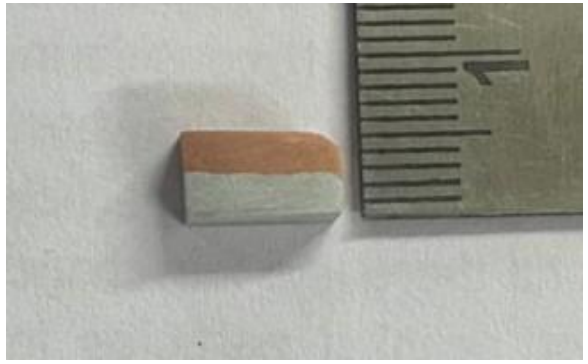
Furnace and Infrared Heating

Furnace Heating

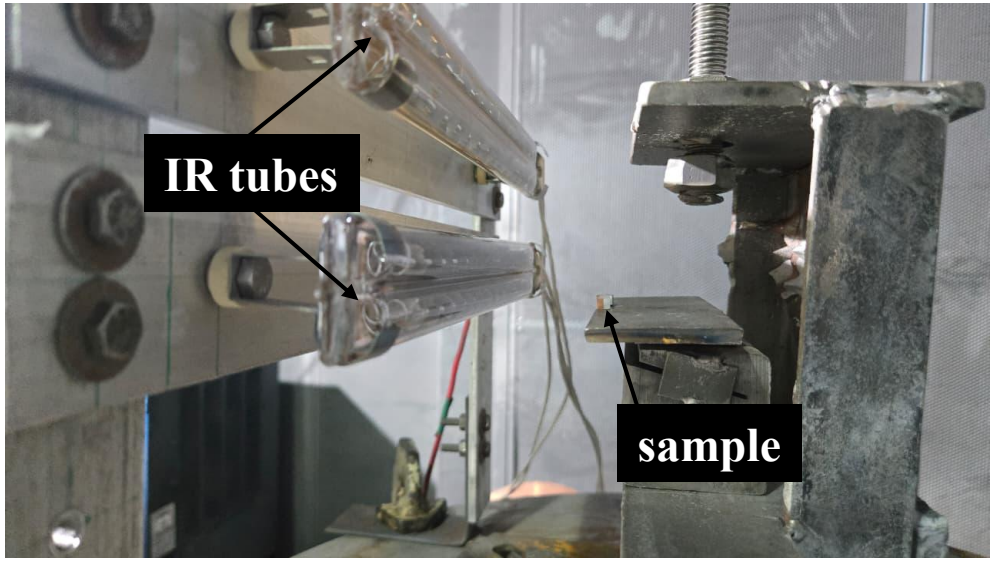




- Total time taken is 13 hours.



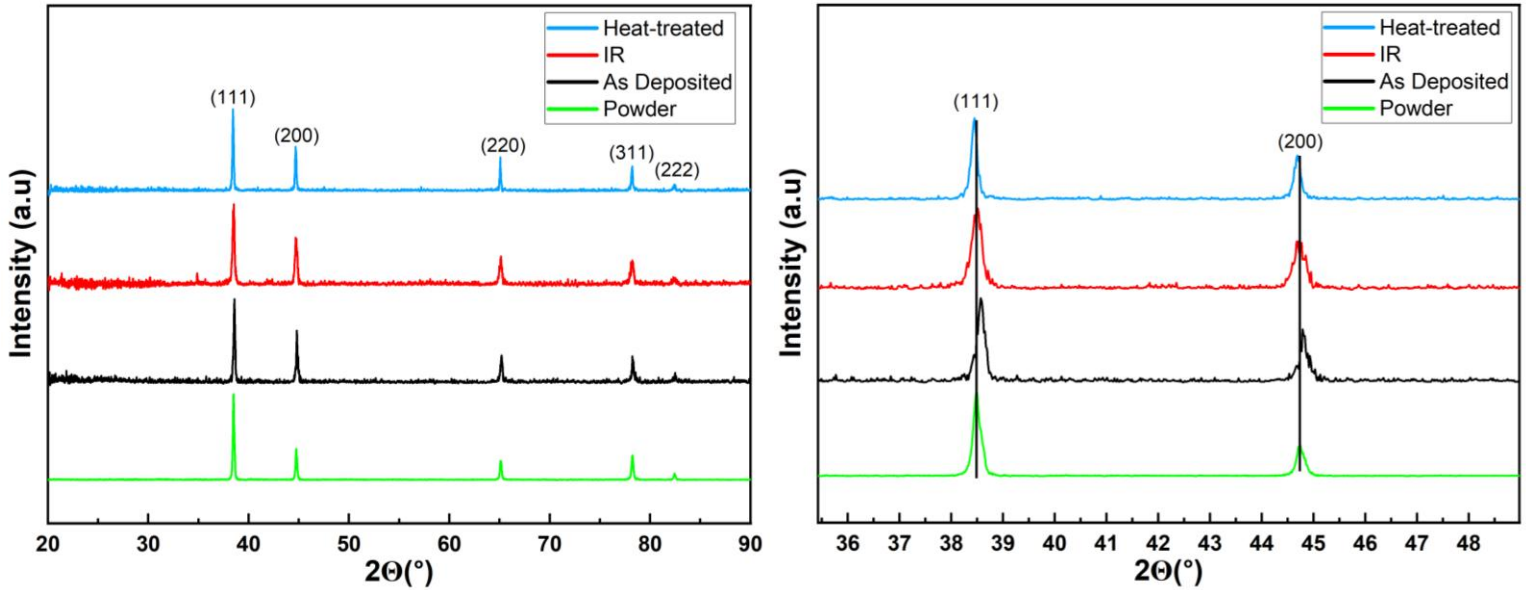
IR Heating

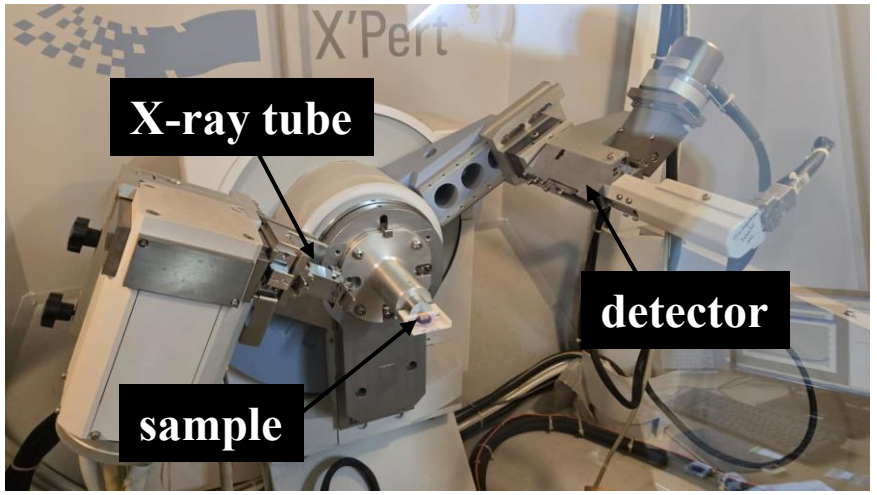


- Heated 3 minutes per each side of the deposit.
- Total time taken in 6 minutes per sample.

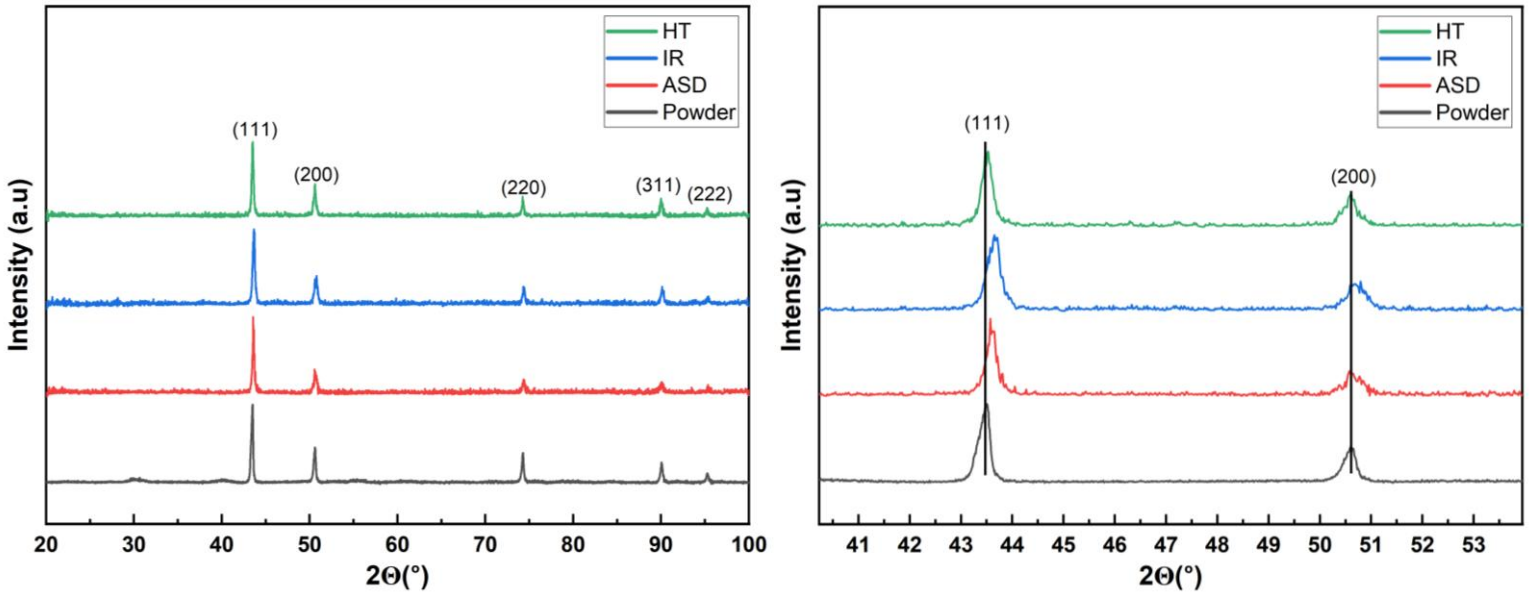
Phase Analysis

➢ Aluminium XRD





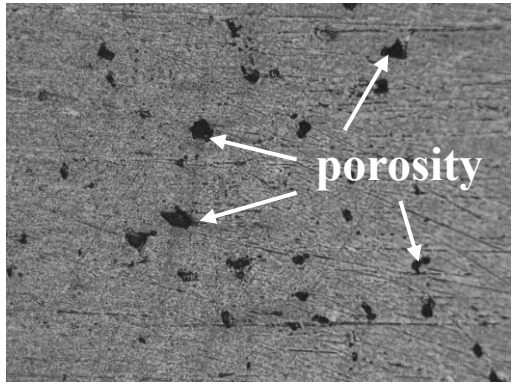
➢ Copper XRD



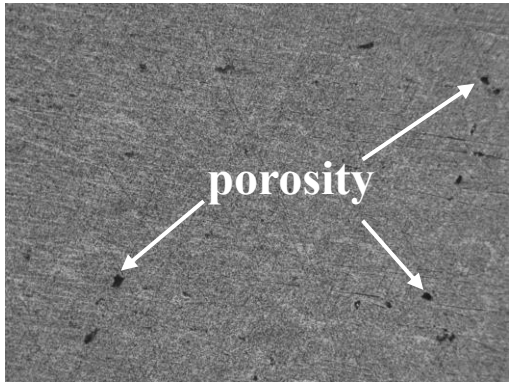
- Peak broadening in XRD represents Microstrain.
- Peak shifting in XRD represents tensile/compressive strain.

Porosity Analysis

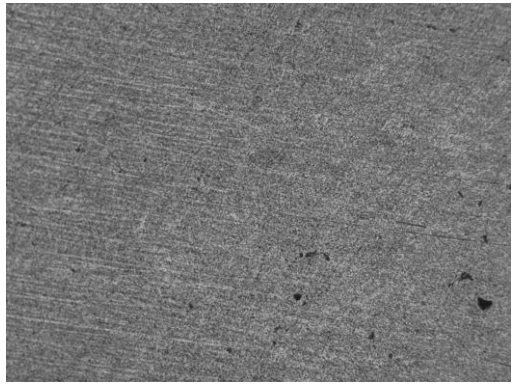
- **Aluminium**



ASD ~2.524±1.15



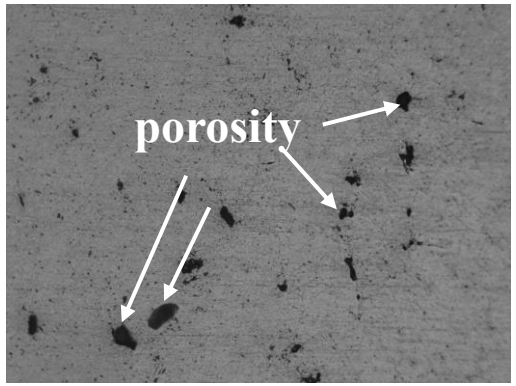
IR ~0.58±0.21



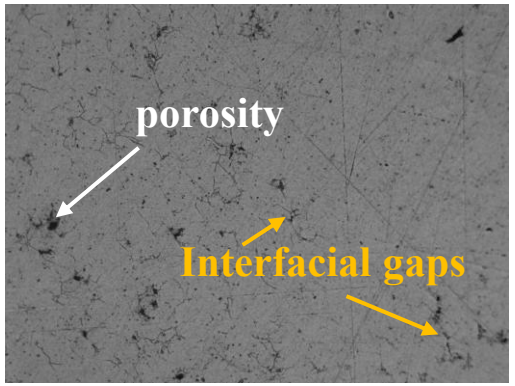
HT ~0.17±0.09

- Due to mismatch of Thermal expansion coefficients Interfacial gaps occurs
- Post-heat treatment can also cause uneven local plastic deformation which leads to delamination.

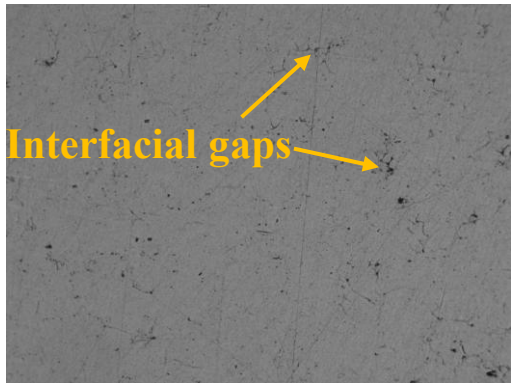
- **Copper**



ASD ~3.143±0.58

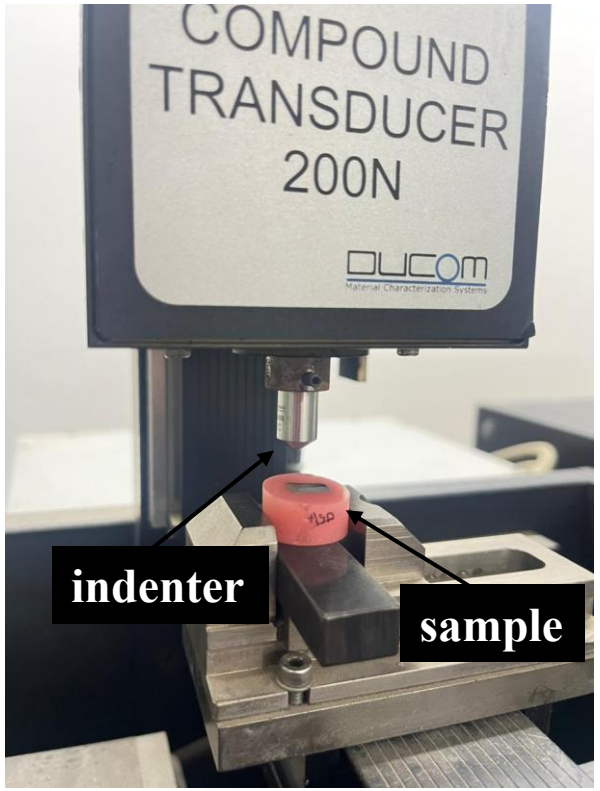


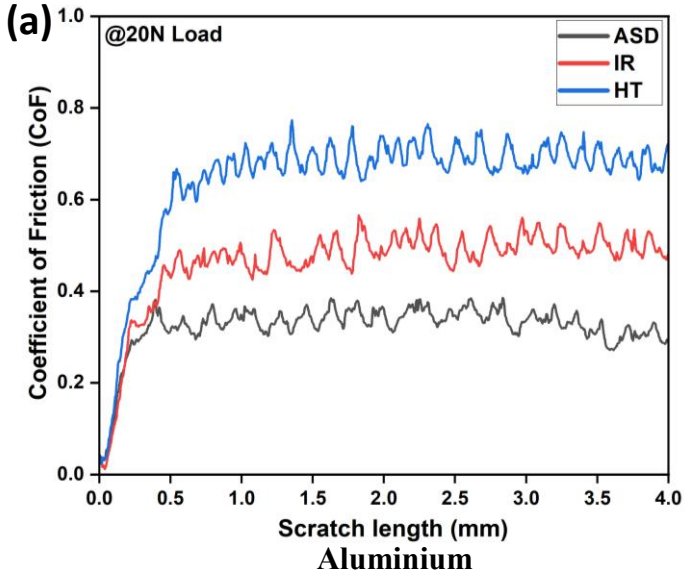
IR ~0.4±0.18

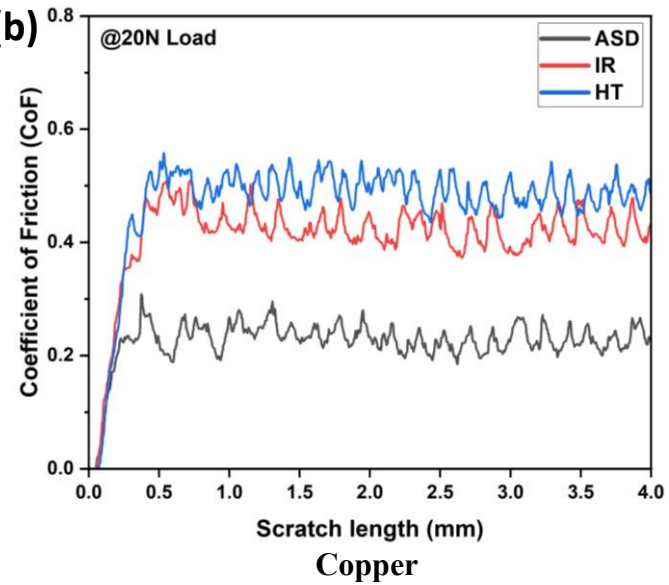


HT ~0.21±0.12

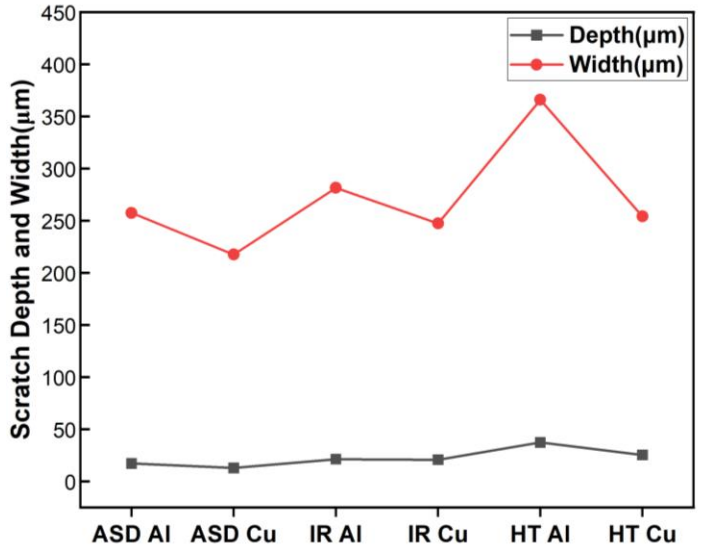
Scratch Analysis



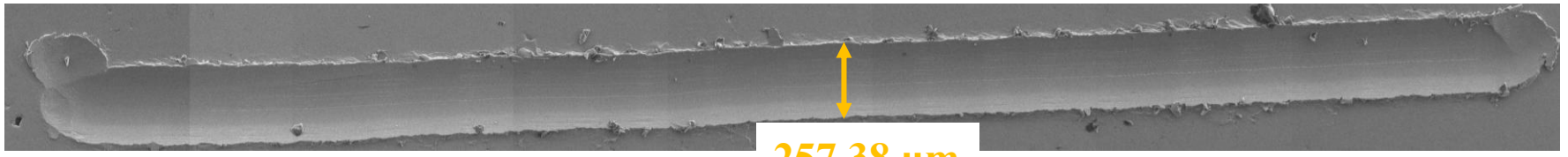


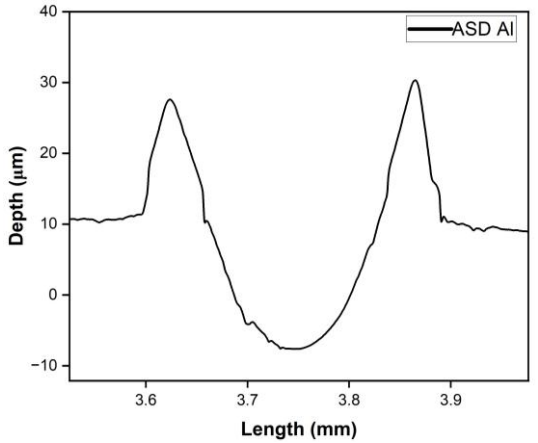


- The scratch width shows exponential growth from ASD to Heat Treatment in Aluminium.
- Meanwhile Copper doesn't show much change among the processes.

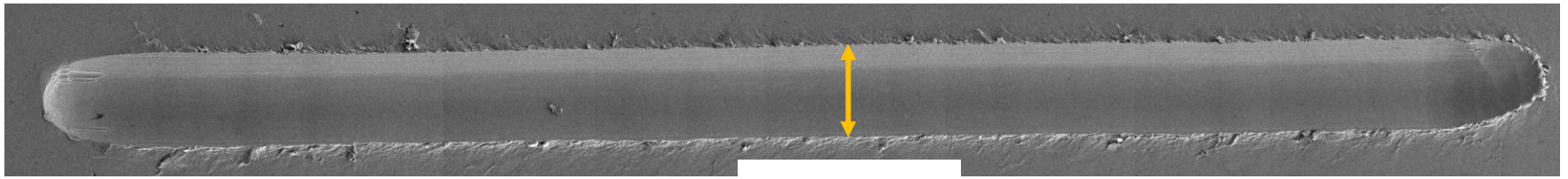


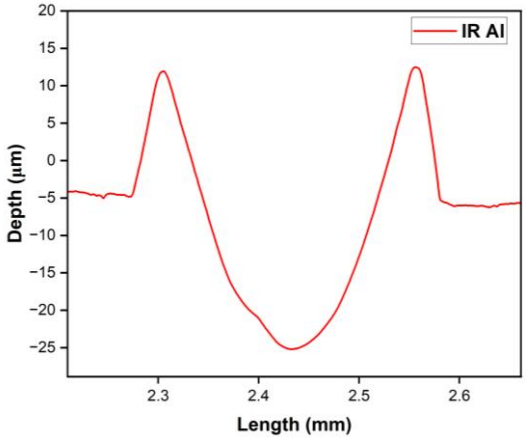
➢ ASD Aluminium



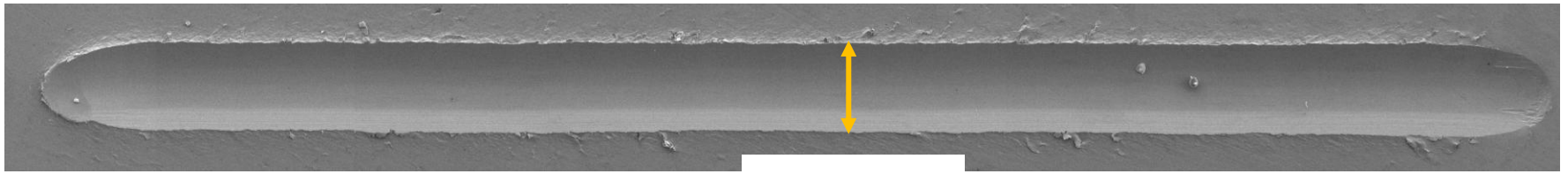


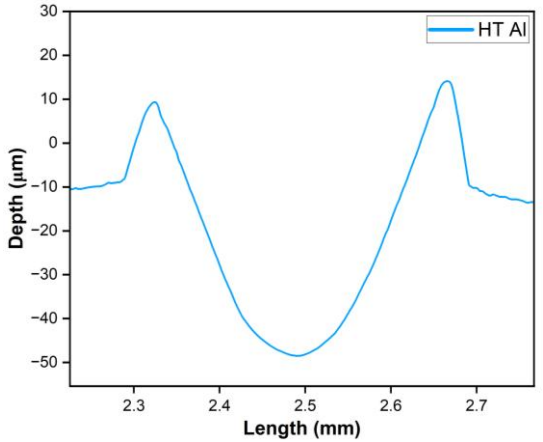
➢ IR Aluminium



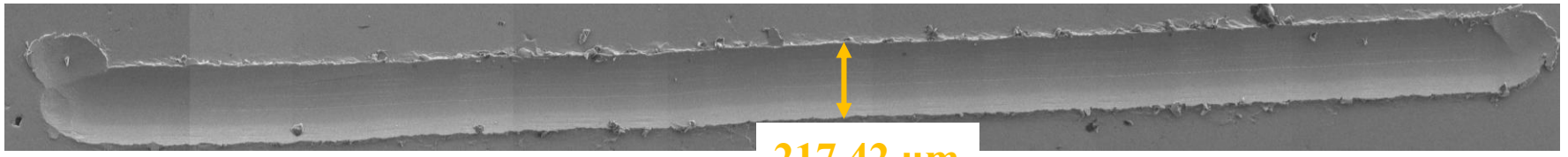


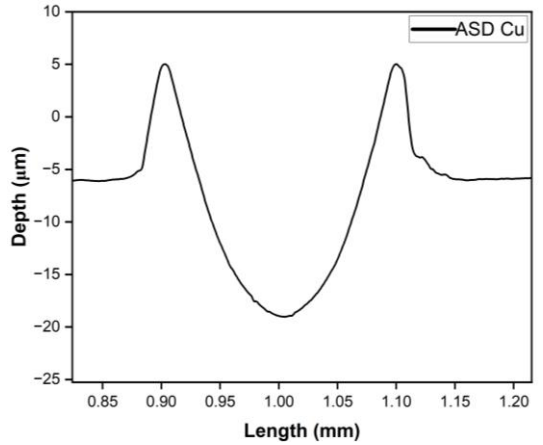
➢ HT Aluminium



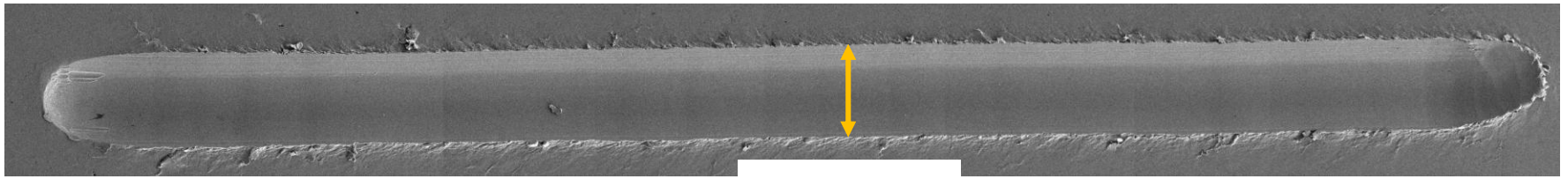


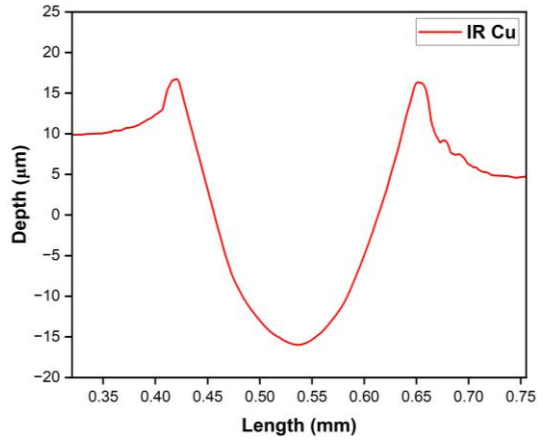
➢ ASD Copper



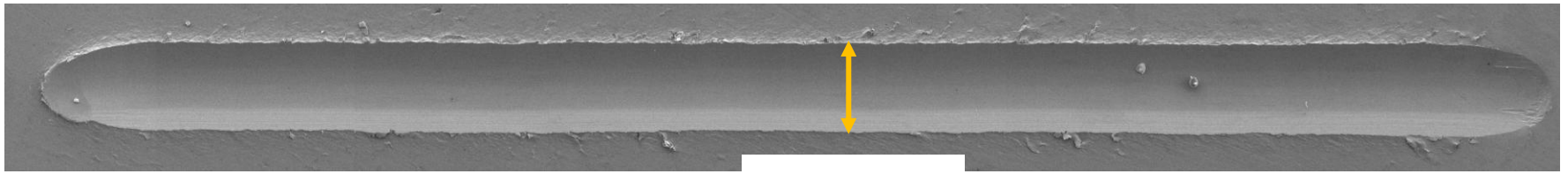


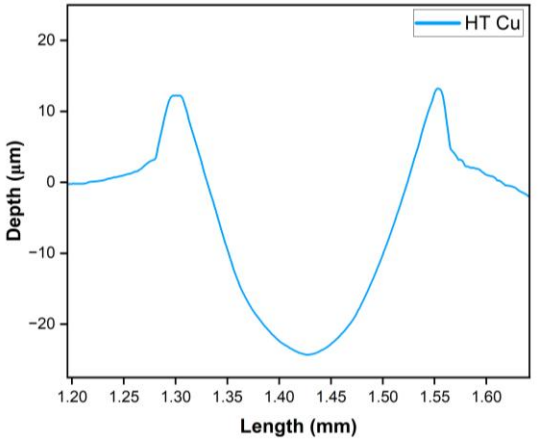
➢ IR Copper



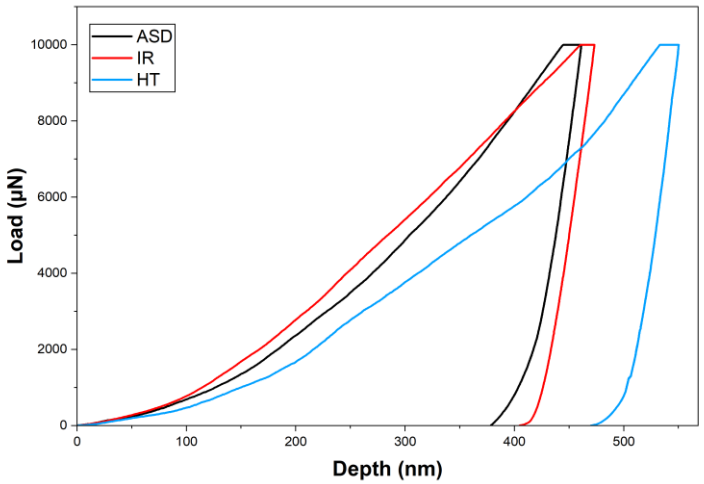


➢ HT Copper



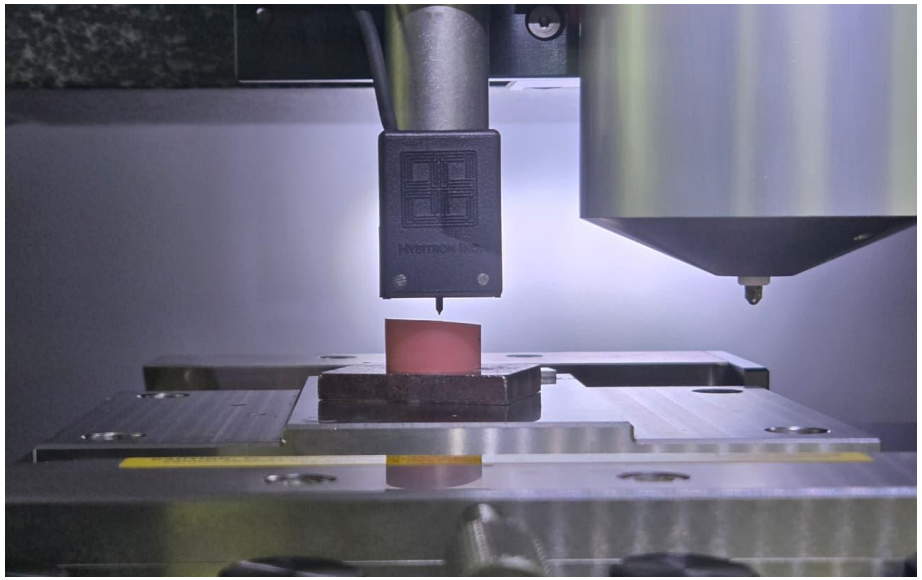


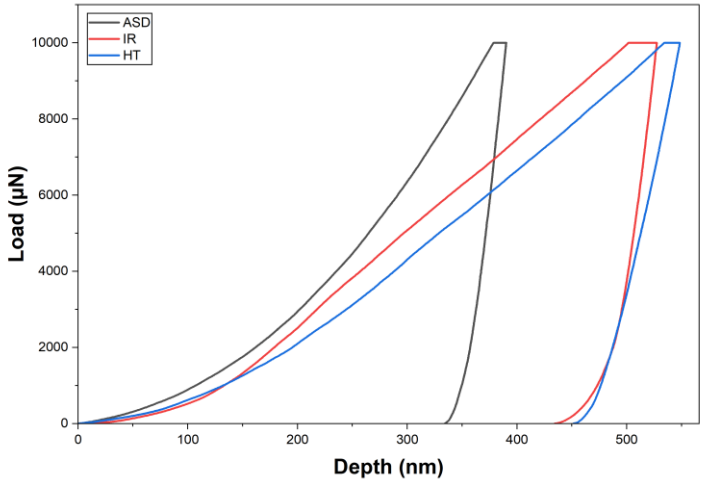
Nano Indentation



(Aluminium)

Hardness Values :
ASD = 2.56 Gpa
IR = 1.47 Gpa
HT = 1.41 Gpa





(Copper)

Hardness Values :
ASD = 1.86 GPa
IR = 1.80 Gpa
HT = 1.28 Gpa

Key Learnings and Conclusion

- Furnace heat treatment requires significantly more time compared to IR heating.
- Porosity is considerably lower in furnace-treated samples than in IR-heated samples.
- XRD analysis shows that peak positions return closer to their original state in furnace heat treatment, whereas in IR heating, full peak recovery is not achieved.
- Scratch test results indicate that both the depth and width increase progressively from as-deposited (ASD) to IR and then to furnace heat-treated samples, suggesting a reduction in brittleness after heating.
- Nanoindentation hardness and load–depth analysis indicate a reduction in brittleness in both aluminum and copper after heating. However, IR and furnace treatments yield nearly similar hardness values, likely due to the higher thermal conductivity of copper influencing the IR heating response.

Thank You

Questions and Discussion

References

- Evolution of microstructure and mechanical properties of cold spray additive manufactured aluminum deposit on copper substrate https://www.sciencedirect.com/science/article/pii/S092150932301448X?utm_source=chatgpt.com#abs0015
- Rajpopat, N., Nourian, A., Jennings, T., Sevinsky, A., Boese, S., and Müftü, S., 2026, “Effects of Spray Conditions and Heat Treatments on the Physical and Mechanical Properties of Al6061 Deposits Cold-Sprayed Using N2 Gas,” Journal of Thermal Spray Technology. https://doi.org/10.1007/s11666-026-02189-3.
- The effects of heat treatment on the mechanical properties of cold-sprayed coatings https://www.sciencedirect.com/science/article/pii/S0257897214010056
- The effects of heat treatment on 7075 Al cold spray deposits https://www.sciencedirect.com/science/article/pii/S0257897216310611
- Murray, J. W., Zuccoli, M. V., and Hussain, T., 2018, “Heat Treatment of Cold-Sprayed C355 Al for Repair: Microstructure and Mechanical Properties,” Journal of Thermal Spray Technology, **27**(1–2), pp. 159–168. https://doi.org/10.1007/s11666-017-0665-z.